

SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 5 – Attitude Kinematics (2)

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Task 1: Simulate the kinematics using Euler Angles

Principal moments of Inertia

$I_x = 0.0700\ \text{kg m}^2$

$I_y = 0.0550\ \text{kg m}^2$

$I_z = 0.0250\ \text{kg m}^2$

Initial conditions

[You choose the I.C.s]

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \dfrac{I_y - I_z}{I_x}\,\omega_y\omega_z \\ \dot{\omega}_y = \dfrac{I_z - I_x}{I_y}\,\omega_x\omega_z \\ \dot{\omega}_z = \dfrac{I_x - I_y}{I_z}\,\omega_y\omega_x \end{cases}$$

Kinematics

$$\begin{cases} \dot{\phi} = \dfrac{\omega_z \cos(\psi) - \omega_x \sin(\psi)}{\cos(\vartheta)} \\ \dot{\vartheta} = \omega_x \cos(\psi) + \omega_z \sin(\psi) \\ \dot{\psi} = \omega_y - [\omega_z \cos(\psi) - \omega_x \sin(\psi)]\dfrac{\sin(\vartheta)}{\cos(\vartheta)} \end{cases}$$

Tasks

- Simulate the kinematics by accounting for the presence of singularities
- Build the DCM from the Euler Angles

$$\mathrm{A}_{312} = \begin{bmatrix} \cos\psi\cos\phi - \sin\psi\sin\phi\sin\vartheta & \cos\psi\sin\phi + \sin\psi\cos\phi\sin\vartheta & -\sin\psi\cos\vartheta \\ -\sin\phi\cos\vartheta & \cos\phi\cos\vartheta & \sin\vartheta \\ \sin\psi\cos\phi + \cos\psi\sin\phi\sin\vartheta & \sin\psi\sin\phi - \cos\psi\cos\phi\sin\vartheta & \cos\vartheta\cos\psi \end{bmatrix}$$

Task 2: Simulate the pointing error of a spin-stabilized S/C

Principal moments of Inertia

$I_x = 0.0700\ \text{kg m}^2$
$I_y = 0.0550\ \text{kg m}^2$
$I_z = 0.0250\ \text{kg m}^2$

Initial conditions

$\omega_x(0) = \pm C\ \text{rad/s}$
$\omega_y(0) = 0.1\ \text{rad/s}$
$\omega_z(0) = 0.1\ \text{rad/s}$
$C \in [0.2, 2\pi]\ \text{rad/s}$

Governing Equations

Dynamics

$$\begin{cases} \dot{\omega}_x = \dfrac{I_y - I_z}{I_x}\,\omega_y\omega_z \\ \dot{\omega}_y = \dfrac{I_z - I_x}{I_y}\,\omega_x\omega_z \\ \dot{\omega}_z = \dfrac{I_x - I_y}{I_z}\,\omega_y\omega_x \end{cases}$$

Kinematics

$$\begin{cases} \dot{\phi} = \dfrac{\omega_z \cos(\psi) - \omega_x \sin(\psi)}{\cos(\vartheta)} \\ \dot{\vartheta} = \omega_x \cos(\psi) + \omega_z \sin(\psi) \\ \dot{\psi} = \omega_y - [\omega_z \cos(\psi) - \omega_x \sin(\psi)]\dfrac{\sin(\vartheta)}{\cos(\vartheta)} \end{cases}$$

Tasks

- Compute initial attitude parameters such that, in inertial frame, $\mathbf{h}(0) \parallel [1, 0, 0]^T$
- Simulate the kinematics by accounting for the presence of singularities
- Compute the pointing error with respect to the nominal direction $[1, 0, 0]^T$ in the inertial frame